```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import load_wine
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

# Load Wine dataset
wine = load_wine()
X = wine.data
y = wine.target
print("Original data shape:", X.shape)
print("Number of features:", X.shape[1])
print("Number of classes:", len(np.unique(y)))
```

```
Original data shape: (178, 13)
Number of features: 13
Number of classes: 3
```

```
#Standardize the features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
print("Data standardized successfully.")
```

```
Data standardized successfully.
```

```
# Apply PCA with all components
pca = PCA()
X_pca = pca.fit_transform(X_scaled)
print("PCA applied successfully.")
print("Number of components:", pca.n_components_)
```

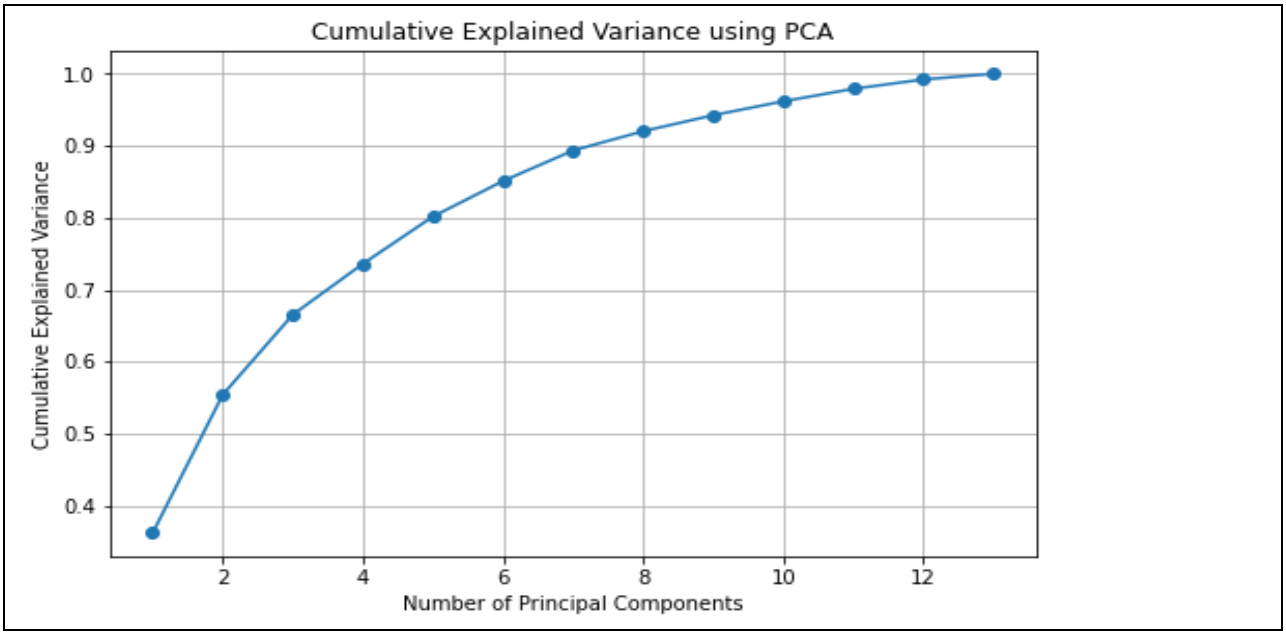
```
PCA applied successfully.
Number of components: 13
```

```
#Examine explained variance
print("Explained variance ratio:")
for i, variance in enumerate(pca.explained_variance_ratio_):
    print("PC", i + 1, ":", round(variance, 4))
```

```
Explained variance ratio:
PC 1 : 0.362
PC 2 : 0.1921
PC 3 : 0.1112
PC 4 : 0.0707
PC 5 : 0.0656
PC 6 : 0.0494
PC 7 : 0.0424
PC 8 : 0.0268
PC 9 : 0.0222
```

```
PC 10 : 0.0193
PC 11 : 0.0174
PC 12 : 0.013
PC 13 : 0.008
```

```
#Plot cumulative explained variance
# Calculate cumulative variance
cumulative_variance = np.cumsum(pca.explained_variance_ratio_)
# Plot
plt.figure(figsize=(8, 5))
plt.plot(
    range(1, len(cumulative_variance) + 1),
    cumulative_variance,
    marker='o'
)
plt.xlabel("Number of Principal Components")
plt.ylabel("Cumulative Explained Variance")
plt.title("Cumulative Explained Variance using PCA")
plt.grid()
plt.show()
```




```
#Reduce data to 2 principal components
# PCA with 2 components
pca_2 = PCA(n_components=2)
X_2D = pca_2.fit_transform(X_scaled)
print("Original dimensions:", X.shape[1])
print("Reduced dimensions:", X_2D.shape[1])
print("Variance preserved:",
      round(pca_2.explained_variance_ratio_.sum(), 4))
```

```
Original dimensions: 13
Reduced dimensions: 2
Variance preserved: 0.5541
```

```
#Visualize the 2D PCA data
plt.figure(figsize=(8, 6))
plt.scatter(
    X_2D[:, 0],
    X_2D[:, 1],
    c=y,
    cmap='viridis'
)
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("Wine Dataset after PCA")

plt.colorbar(label="Wine Class")
plt.grid()
plt.show()
```

